

ESKILSTUNA, Sweden

WMO: 024490

Lat: 59.3825N

Lon: 16.4522E

Elev: 12

StdP: 101.18

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.2	-15.3	-20.4	0.6	-18.7	-16.8	0.9	-14.7	10.1	3.1	8.8	2.8	0.9	210	0.580

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	27.4	18.1	25.5	17.5	23.8	16.7	19.8	24.0	18.7	23.2	17.7	22.2	3.0	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.4	13.3	21.2	17.1	12.3	20.2	15.9	11.3	19.3	56.6	24.0	52.9	23.0	49.8	22.3	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.0	7.7	6.6	DB	-21.8	29.3	5.0	1.6	-25.4	30.4	-28.3	31.3	-31.0	32.2	-34.6	33.3	(4)
(5)			WB	-20.4	20.8	4.8	1.3	-23.8	21.8	-26.6	22.5	-29.3	23.3	-32.7	24.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.6	-3.7	-2.5	1.0	5.5	10.9	14.7	17.5	16.0	11.7	6.5	2.6	-1.4	(6)	
	DBStd	8.23	5.64	5.28	4.22	3.41	3.63	2.99	2.67	2.64	3.13	3.83	3.93	5.52	(7)	
	HDD10.0	1941	424	349	280	138	33	2	0	0	18	121	223	353	(8)	
	HDD18.3	4313	682	582	538	384	232	116	49	81	198	367	473	611	(9)	
	CDD10.0	707	0	0	0	4	60	142	232	187	71	12	1	0	(10)	
	CDD18.3	38	0	0	0	0	1	6	23	9	1	0	0	0	(11)	
	CDH23.3	472	0	0	0	0	22	92	273	84	2	0	0	0	(12)	
	CDH26.7	88	0	0	0	0	1	13	64	11	0	0	0	0	(13)	
(14) Wind	WSAvg	2.9	2.9	3.0	3.2	3.0	2.9	3.0	2.7	2.5	2.7	2.8	2.9	3.0	(14)	
(15) Precipitation	PrecAvg	603	36	33	27	34	42	65	83	76	53	55	53	46	(15)	
	PrecMax	761	78	66	55	100	83	116	196	180	154	123	120	95	(16)	
	PrecMin	532	8	12	3	4	16	19	24	5	8	24	15	12	(17)	
	PrecStd	54	17	13	10	24	19	26	38	41	34	25	29	22	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.7	8.5	13.7	20.1	25.5	28.3	30.7	27.9	23.1	16.7	12.1	10.4	(19)	
		MCWB	5.3	6.4	8.2	11.0	15.1	17.6	18.5	19.2	16.5	13.3	10.7	8.7	(20)	
	2%	DB	5.3	6.6	11.0	17.3	23.5	26.0	28.5	25.6	20.8	14.9	10.5	8.3	(21)	
		MCWB	4.2	4.6	6.5	9.9	14.4	17.0	18.4	18.5	15.6	12.5	9.5	6.9	(22)	
	5%	DB	4.1	5.4	8.7	14.9	21.1	23.7	26.4	23.7	18.8	13.7	9.2	6.4	(23)	
		MCWB	3.3	4.1	5.4	8.9	13.5	15.9	17.9	17.5	14.7	11.7	8.2	5.2	(24)	
	10%	DB	3.0	4.2	6.7	12.5	18.6	21.7	24.4	22.0	17.2	12.4	8.0	5.1	(25)	
		MCWB	2.2	3.0	4.1	7.4	12.3	14.9	17.5	16.5	14.0	10.8	7.1	4.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.7	7.2	9.1	11.9	17.1	19.2	21.5	21.2	17.5	14.3	11.1	9.1	(27)	
		MCDB	6.4	7.9	12.6	18.0	23.6	25.2	25.4	25.1	21.7	15.8	11.8	10.2	(28)	
	2%	WB	4.4	5.2	7.1	10.5	15.5	18.0	20.1	19.8	16.3	13.0	9.8	7.2	(29)	
		MCDB	5.0	6.0	10.4	15.8	20.8	23.4	24.3	23.6	19.6	14.6	10.5	8.2	(30)	
	5%	WB	3.5	4.2	5.9	9.5	14.1	16.8	19.0	18.5	15.4	12.1	8.6	5.5	(31)	
		MCDB	4.1	5.2	8.3	14.2	19.7	22.4	23.5	22.2	18.0	13.4	9.2	6.3	(32)	
	10%	WB	2.3	3.1	4.7	8.2	12.9	15.7	18.0	17.3	14.5	11.2	7.4	4.2	(33)	
		MCDB	3.0	4.2	6.5	12.0	18.3	20.4	23.0	20.8	16.9	12.4	8.1	5.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.5	6.3	7.8	10.8	11.4	10.9	10.7	10.3	9.3	7.4	4.6	5.1	(35)	
		MCDBR	5.1	5.5	10.4	15.5	15.8	15.5	15.2	13.3	11.4	7.6	6.0	5.2	(36)	
	5% WB	MCWBR	4.6	4.4	6.9	9.3	8.7	7.8	7.0	6.9	6.8	5.5	5.4	5.1	(37)	
		MCWBR	4.7	5.0	9.7	13.6	13.7	12.9	12.1	11.6	10.3	7.0	5.9	5.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.276	0.317	0.338	0.375	0.371	0.362	0.387	0.379	0.345	0.334	0.303	0.260	(40)	
		taud	2.209	2.275	2.353	2.316	2.393	2.441	2.391	2.408	2.491	2.422	2.269	2.140	(41)	
	Ebn at Noon		523	689	798	825	858	872	837	815	790	672	483	393	(42)	
		Edh at Noon	43	69	87	107	106	103	106	97	76	60	42	32	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.32	0.95	2.46	3.94	5.11	5.72	5.23	4.14	2.73	1.22	0.40	0.21	(44)	
	RadStd		0.05	0.14	0.30	0.39	0.46	0.34	0.52	0.41	0.31	0.20	0.09	0.02	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-121	-147	N/A	N/A

Nomenclature: See separate page